ZAGROS – Project Definition & Elicitation Plan

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Task 1: Project definition

1. Project title and problem statement

Title: ZAGROS: Erbil Terminal Air Traffic Control System

Problem statement: Controllers at Erbil International Airport (ORER) handle a mixed flow of arrivals, departures, overflights and ground movements, often while dealing with sudden weather changes and regional airspace disruptions. Today the traffic picture, clearances and flight strips are spread across separate screens and paper records. That makes it hard to see conflicts early, react quickly when the runway in use changes, and keep a full record of every instruction. Mistakes or delays in any of these areas directly affect flight safety and airport capacity.

2. Measurable objectives

1. Display every aircraft within 60 NM of Erbil, airborne or on the ground, with its position updated at least once per second.
2. Alert the controller to any predicted loss of separation (less than 3 NM and 1,000 ft) at least 90 seconds before it happens, in 100% of test scenarios.
3. Let a controller issue any clearance (heading, altitude, landing, take-off, taxi) in 2 clicks or fewer, with each clearance logged with time and controller ID.
4. Detect a runway incursion and alert the tower within 2 seconds.

3. Scope

In scope	Out of scope
A live radar map of the Erbil terminal area (60 NM, surface to FL245)	Real radar hardware or a live ADS-B feed (we use a simulator)
Clearances for arrivals and departures (vectors, landing, take-off, holding)	En-route control above FL245 (that belongs to Kirkuk ACC)
Ground movement: stands, pushback, taxi routes, holding points and stop bars	Official certification for real air traffic operations

In scope	Out of scope
Safety alerts: aircraft conflicts, terrain (Zagros mountains), runway incursion and weather	Airline systems such as ticketing, check-in and baggage
Electronic flight strips and a radio log of all instructions	
Runway selection based on the wind (runway 18 or 36)	
A training mode with scenarios (emergency, storm, wind shift)	

4. Stakeholders and their needs

Stakeholder	What they need
Approach controller	A clear picture of all airborne traffic and early conflict warnings so they can space arrivals safely
Tower controller	Runway status at a glance and protection against clearing two aircraft onto the runway at once
Ground controller	Taxi routes and stand availability so aircraft move on the ground without delays or collisions
ATC supervisor	Control over which runway is in use, plus a full record of events for incident review
Pilots / airlines	Clear, timely instructions and minimal delays in holding or on the ground
Airport operations	Accurate arrival and departure times to manage gates and staff
Instructor / trainee controllers	Realistic training scenarios that can be repeated safely

5. Feasibility

- **Technical:** Yes. It can be built with React, Canvas, Node.js and PostgreSQL/PostGIS, and our working prototype already shows the radar, alerts and ground operations running.

- **Economic:** Yes. All the tools are free or covered by student plans (GitHub, Vercel, Supabase), so the only real cost is the team's time.
- **Operational:** Yes for training use. The interface follows real ICAO phraseology and controller workflows, but it would need certification before any real operational use.

Task 2: Elicitation plan and role-play

1. Elicitation techniques

- **Interviews:** ATC work is specialised and safety-critical, so we need to talk directly to controllers to understand how they think, which decisions they make and what goes wrong. You can't get that from a questionnaire.
- **Observation:** Controllers often do things automatically and forget to mention them in an interview, like scanning the radar in a set pattern or using hand-written notes. Watching a shift (or recorded ATC audio and videos) shows the real workflow.
- **Prototyping:** Controllers can react much better to something they can see. Showing them our ZAGROS radar prototype lets them point out exactly what's missing or confusing.

Check these names match the techniques listed in Part 1.IV and swap if needed.

2. Interview questions (for an Erbil approach or tower controller)

1. **(Springboard)** Walk me through the last busy hour you worked at Erbil. What was happening, and what did you do?
2. **(Open)** When the wind changes and you have to switch from runway 36 to 18, what steps do you go through?
3. **(Closed)** Do you currently use paper flight strips, electronic strips, or both?
4. **(Open)** Tell me about the last time you noticed a possible conflict between two aircraft. How did you first spot it?
5. **(Closed)** How many seconds of warning would you need before a conflict alert is useful rather than annoying?
6. **(Open)** What usually causes delays for aircraft on the ground, between the gate and the runway?
7. **(Closed)** When an aircraft declares an emergency, do you need the screen to change automatically, or do you prefer to control that yourself?
8. **(Open)** If you could fix one thing about the screens or tools you use today, what would it be, and why?

3. Role-play recording sheet

Fill this in while interviewing the other team's client.

#	New need discovered	Who needs it	Priority (M/S/C)
1			
2			
3			
4			

Example new needs for ZAGROS (from playing the controller):

- The controller wants alert sounds that can be muted for a short time, because constant alarms are distracting during busy periods.
- Approach controllers want to see the estimated landing time for each arrival, not just its distance.
- Ground controllers need to know when a stand is blocked for maintenance, not just whether a plane is parked there.
- Supervisors want a shift handover summary showing which aircraft are being handled and which alerts are still open.
- Controllers want the radio log to be searchable by callsign, to check quickly what they told a pilot earlier.

Tip for the role-play: when you play the controller for the other team, answer as if you really work at Erbil Tower. Mention real details like runway 18/36, the Zagros terrain, Turkish and Qatar flights, and thunderstorms in spring.